

# Ashray Kapuria

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## Education

### Drexel University - School of Biomedical Engineering, Science and Health Systems

Master of Science, Biomedical Sciences

June 2025

### University of Oregon - College of Arts and Sciences

Bachelor of Science, Biology (Emphasis: Ecology & Evolution), Minor in Chemistry

January 2022

## Profile

Biomedical Sciences graduate with hands-on experience in clinical research, statistical modeling, and healthcare data analysis. Co-author of multiple nationally presented studies, including a first-author project accepted by ESPU. Skilled in deriving insights from complex datasets and translating findings into actionable recommendations.

## Work Experience

### Children's Hospital of Philadelphia (CHOP) - Research Associate

Philadelphia, PA | December 2023 - Present

- Designed and executed multivariable statistical analyses on 213 pediatric hypospadias cases to evaluate DIG vs. TD surgical outcomes; results informed clinical protocols and led to implementation of a new EMR alert for high-risk anatomy cases
- First author of study on preoperative testosterone and BMI; performed z-score normalization and between-group comparisons to identify correlations in growth response; accepted for presentation at the European Society of Pediatric Urologists (ESPU 2025, Vienna)
- Co-authored abstracts on BMG reconstruction and complication predictors; presented findings at SPU and AUA conferences based on demographic stratification, outcome frequencies, and complication modeling
- Built and managed REDCap datasets from >2000 EMR records; extracted variables including chordee, glans groove morphology, and urethroplasty length; aligned statistical output with clinical endpoints
- Captured and organized pre-, intra-, and post-operative imagery to support AI complication prediction model development (ARCUS project); collaborated with surgeons and data science teams to structure image files and metadata
- Co-led retrospective review of 3 legacy datasets; developed new charting frameworks, integrated missing anatomical markers, and trained new interns on REDCap and literature curation
- Contributed to surgical ergonomics study using wearable posture monitors; led intraoperative data capture and analysis pipeline for ongoing human factors research

### Kaiser Permanente – Research Intern

Baldwin Park, CA | September 2022 – October 2023

- Co-authored study on 9-year ambulatory arthroplasty outcomes; presented at the American Society of Anesthesiologists Annual Meeting (Oct 2023, San Francisco)
- Conducted statistical analysis (regression, descriptive metrics) on EMR data from total hip and knee replacement patients to identify recovery trends and complication predictors
- Structured and cleaned multi-year clinical datasets in collaboration with bioinformatics team; supported modeling of longitudinal surgical outcomes
- Drafted and edited results and methods sections for peer-reviewed manuscript; aligned statistical output with operational benchmarks
- Contributed to secondary analysis of ventilatory performance in intubated patients using algorithmic thresholds and lung expansion data

### Orange County Orthopedic Group – Medical Intern

Anaheim Hills, CA | Summers 2018–2020

- Shadowed orthopedic and pain management specialists in outpatient procedures
- Supported clinical workflows including patient prep, x-ray setup, and hygienic protocols
- Observed corticosteroid injections and cast removals across diverse patient populations

## SKILLS

**Technical:** MATLAB, R, Python, Java, SQL, Qlik Sense, REDCap, EMR (EPIC), Excel, Google Suite, Microsoft Office

**Analytical:** Statistical modeling, hypothesis testing, data transformation, longitudinal analysis, literature review

**Professional:** Scientific writing, presentation development, cross-functional collaboration, mentoring, project coordination